Plasma Surface Modification of Polyethylene (PE) for Reduced Biofilm Formation

BMEN 343 Biomedical Engineering Materials Laboratory Design Project

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Introduction

Biofilms are a major concern for medical implants, as they are responsible for approximately 65% of microbial infections and 80% of chronic infections [1]. These infections occur when bacteria adhere to implant surfaces and proliferate, forming a protective extracellular matrix that enables persistence and resistance to immune responses and antibiotics. Polyethylene (PE) is a material commonly used for joint implants but is susceptible to biofilm formation due to its high hydrophobicity, which promotes protein adsorption and bacterial attachment [2]. To reduce biofilm formation on implant surfaces, plasma treatment can be used to modify the surface and increase hydrophilicity [3]. The goal of this study is to evaluate whether plasma surface modification reduces biofilm formation by decreasing bacterial attachment and proliferation.

Current Market Solutions

Antibacterial Biomaterial Coatings

Current strategies to prevent implant-associated biofilm formation include antibacterial coatings applied to medical devices. Active coatings, such as antibiotics, silver, or antimicrobial peptides, function by directly killing or inhibiting bacterial growth, while passive coatings, including hydrogels and hyaluronic acid, reduce bacterial adhesion to the surface. Although these methods can be effective, they are often limited by a finite duration of activity, potential cytotoxic effects, and the risk of promoting antibiotic resistance [4].

PEGylation (Hydrophilic Surface Coating)

Another commonly used approach is PEGylation, which involves coating implant surfaces with polyethylene glycol (PEG) to create a hydrophilic barrier. PEG binds water molecules to form a hydration layer that reduces protein adsorption and prevents bacterial attachment, thereby limiting the initial stages of biofilm formation. While PEG is highly biocompatible and effective at reducing adhesion, its coatings can degrade over time and do not actively eliminate bacteria[4].

Methods

Three samples of PE were plasma treated to modify the surface and three were left untreated.

FTIR Spectroscopy

Fourier Transform Infrared (FTIR) spectroscopy is used to identify the functional groups of the sample. The goal was to identify if plasma treatment added polar groups to treated samples which would reduce biofilms [6].

Static Contact Angle

To assess the difference in hydrophilicity, a static contact angle was measured. If the contact angle of the treated or untreated sample group was less than 90 degrees, the sample group was inferred to be hydrophilic [7].

Dynamic Contact Angle

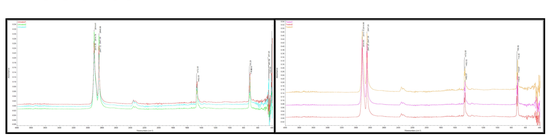
To study change in contact angle over time, dynamic contact angle was measured from 0 to 30 seconds at intervals of 5 seconds [7].

Bradford Assay

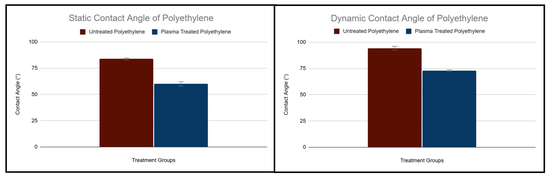
Bradford's reagent was used to study protein adsorption of sample groups. Stronger dye retention would translate to greater hydrophobicity [8]. A standard dilution curve was graphed using known concentration values as shown in Figure 5.

All quantifiable measurements were conducted for treated and untreated groups with results reported using mean ± standard deviation. Statistical significance was evaluated using a t-test with a significance level of α=0.05.

Results



Figures 1 & 2. Untreated and Plasma Treated Polyethylene FTIR Spectra



Figures 3 & 4 Bar Chart Analysis of Static and Dynamic Contact Angle of Polyethylene

Table 1. Static Contact Angle of Untreated and Plasma-Treated Polyethylene

Untreated (°)	Plasma Treatment (°)	P-value (α = 0.05)	Significant?
83.820 ± 0.873	59.993 ± 2.066	0.041	Yes

Table 2. Dynamic Contact Angle of Untreated and Plasma-Treated Polyethylene

Untreated (°)	Plasma Treatment (°)	P-value (α = 0.05)	Significant?
94.258 ± 1.836	73.066 ± 0.577	0.021	Yes

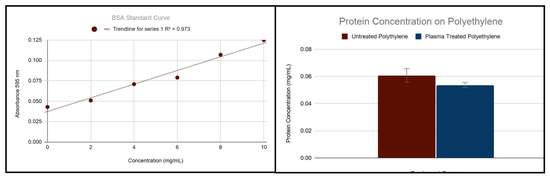


Figure 5 & 6. Bovine Serum Albumin (BSA) Standard Curve and Protein Concentration for Bradford Assay

Table 3. Bradford Assay Absorbance at 595 nm of Untreated and Plasma-Treated Polyethylene

Untreated	Plasma Treated	Percent Change (%)	P-value (α = 0.05)	Significant?
0.061 ± 0.005	0.054 ± 0.001	-11.689 ± 1.0068	0.006	Yes

Discussion & Conclusion

The comparison of the FTIR before and after plasma treatment shows the spectra of the PE remains unchanged [6]. Plasma cleaning is a surface technique while FTIR analyzes the bulk of the material. There is slightly increased noise at the end of the spectra as a result of the surface of the polyethylene becoming more rough from plasma cleaning.

The static contact angle results showed that there was a decrease in contact angle on the plasma treated samples, suggesting that the PE became more hydrophilic. Both untreated and treated samples exhibited contact angles below 90°, classifying them as hydrophilic [7], statistical analysis confirmed that the decrease after plasma treatment was significant.

The dynamic contact angle results aligned with our static contact angle trend, and remained significant across our trials. In dynamic conditions, our untreated sample was greater than 90° meaning it is hydrophobic, while plasma treated remained hydrophilic [7]. The higher dynamic angles compared to static angles are attributed to surface hysteresis.

Bradford assay results indicated a significant difference in protein concentration between treated and untreated samples. However, plasma treatment did not completely eliminate protein adhesion, suggesting that the increase in hydrophilicity alone may not be sufficient to fully prevent protein–surface interactions. Previous studies have shown that modest increases in hydrophilicity are often insufficient to resist protein adsorption, as long-term resistance to biofouling depends on more stable and durable surface modifications [9].

Overall, plasma treatment increased PE surface hydrophilicity under both static and dynamic conditions but was not sufficient to fully prevent protein adhesion, suggesting that additional strategies may be needed to reduce biofouling.

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